

RoutelQ

Evidence-driven inference routing for open models

Project lead: Hossam Elshahaby

RoutelQ is an AI inference control plane that chooses the least expensive and fastest eligible model that can satisfy a workload's required quality. Nebius Token Factory remains the inference layer; RoutelQ adds routing, measurement, benchmark evidence, cost/latency visibility, and an adaptive feedback loop.

Problem

AI applications commonly route heterogeneous workloads through one high-capability model. That hides the economic difference between simple classification or extraction and genuinely difficult reasoning. Open/open-weight models can still incur hosted inference cost because serving consumes GPU infrastructure, input tokens, output tokens, and latency.

Solution

RoutelQ performs capability gating, workload classification, complexity analysis and evidence-aware model selection. It chooses a routing profile (Fast, Balanced or Deep) and then an exact eligible Nebius model. The interface explains why the model was selected and exposes measured latency, token use, verified pricing when available, and benchmark evidence.

Model economics

Inference is executed through Nebius Token Factory. For the hackathon MVP, RoutelQ can enrich Nebius-discovered models with upstream-provider pricing referenced from Requesty's Nebius catalog, while keeping provenance visible. Cost is calculated from measured input/output tokens and the matched per-million-token rates. Missing or ambiguous model matches remain unavailable rather than guessed.

Evidence and benchmarks

BANKING77 measures banking intent-classification accuracy and GSM8K measures mathematical reasoning correctness. Fixed Fast-only and Deep-only control strategies are compared with RoutelQ Auto, which chooses a model before seeing ground truth. Results feed quality-retained, latency, cost and Deep-calls-avoided metrics.

Trade-off explorer

RoutelQ visualizes Quality vs Cost, Quality vs Latency, Accuracy Ranking and Token Efficiency. The efficient frontier highlights models that are not dominated by another measured model with equal-or-better quality and equal-or-lower cost.

Adaptive routing evidence

Users can force another eligible model and compare it with RoutelQ's recommendation. A preference for the alternative becomes bounded, workload-specific routing evidence. RoutelQ learns its routing policy; it does not claim to train or fine-tune the underlying LLM.

Freshness and documents

Tavily is used only when a request requires current external information. Document-only summarization, extraction or risk analysis should use the uploaded document directly. Freshness is determined from user

intent, avoiding unnecessary retrieval latency and noise.

Customer and impact

The demo customer, FinAssist AI, represents a B2B fintech SaaS with mixed support, research and operations workloads. RoutelQ helps prevent simple workloads from consuming unnecessary Deep inference while preserving escalation for difficult reasoning. The business value is measured as quality retained, cost avoided, latency change, over-provisioning avoided and necessary escalations.

Architecture

Customer App → RoutelQ capability/task/complexity/freshness analysis → model eligibility and scoring → Nebius Token Factory → response telemetry → benchmark/runtime evidence → adaptive routing evidence. Pricing reference and provenance remain separate from inference.

Demo flow

1) Submit a simple request and inspect Auto's exact model recommendation. 2) View the metric strip and model trade-off position. 3) Force another eligible model and compare. 4) Record which result is preferred. 5) Run BANKING77 or GSM8K. 6) Inspect Quality vs Cost/Latency and Deep calls avoided. 7) Open Business Impact to translate measured routing into inference economics.

Current scope

The MVP intentionally avoids authentication, billing, custom hosting, fine-tuning, multi-provider inference, complex RAG, avatars and expensive 3D visualization. The goal is one credible, measurable routing loop built around Nebius Token Factory.

Tagline: Fast when sufficient. Deep when necessary.

Pricing reference used by the MVP: Requesty Nebius model catalog. Inference provider: Nebius Token Factory. RoutelQ should always display provenance and timestamps for externally sourced pricing.